SHUFAN SHAHI

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About

AI/ML engineer and CSE undergraduate (IUT, 2026) who takes models from data analysis and experimentation to production APIs. Currently an AI Engineer at GLiTch LABS, building, evaluating, and deploying ML and LLM systems, including RAG with embeddings and reranking, search and ranking, forecasting, customer segmentation, risk alerting, and GA-based optimization. Strong in Python, PyTorch, TensorFlow, Pandas, NumPy, and SQL, with research experience in multimodal learning, NLP, and computer vision, and a top-5 finish in an international anomaly-detection competition. Comfortable with REST services, Docker, Git, and Linux; works independently and with backend and product teams. Based in Dhaka and available for full-time, on-site work.

Education

Islamic University of Technology **2022–2026**
B.Sc. in Computer Science and Engineering **CGPA: 3.58**
Coursework: Algorithms, Databases, Networks, Machine Learning, and Deep Learning.

Chuadanga Govt. College **2021**
Higher Secondary Certificate (HSC) **GPA: 5.00/5.00**

Experience

AI Engineer **2025–Present**
GLiTch LABS

- Design, train, fine-tune, and evaluate ML and LLM-based features in Python (PyTorch, TensorFlow), taking prototypes through experimentation to production-ready services.
- Build RAG and retrieval pipelines with embeddings, vector databases, and reranking, and serve models through backend REST APIs integrated with client applications.
- Prepare datasets, run feature engineering and data analysis with Pandas, NumPy, and SQL, and debug model and pipeline issues with backend engineers and product teams.

Intern **2025**
Battery Low Interactive Ltd.

- Learned Flutter development, REST API integration, and applications of fine-tuned ML models.

Senior Executive **2025–2026**
Project Aero

- Contributed to drone R&D by applying AI-based object tracking for international UAV competitions.

Researcher **2025–2026**
IUT Computer Vision Lab

- Researched multimodal AI methods to improve integration and mitigate modality imbalance.

Student Researcher **2026**
Elite Research Lab

- Support research projects through literature reviews, experiments, and analysis.

Projects

Biztrack - Business Analytics | *Next.js, Node.js, PostgreSQL* GitHub Demo
- Built full-stack AI modules for forecasting, customer segmentation, and risk alerts on transactional data, backed by PostgreSQL and served through APIs.

Postfolio - Professional Networking Platform | *Spring Boot, Next.js, Docker, RabbitMQ, Stripe* GitHub Demo
- Built a containerized full-stack platform with Spring Boot backend services, RabbitMQ messaging, and Stripe payments.
- Built an AI matching engine that ranks candidates against job postings to surface the best-fit hires, alongside portfolio and CV creation, job search, and AI interview preparation.

DocuMind - Document Q&A | *Python, LangChain, LlamaIndex, ChromaDB, Ollama* GitHub
- Built a local RAG system for documents with cited answers and hallucination guardrails.
- Compared 4 chunking methods and 2 embedding models; added hybrid BM25 and reranking.

Benhallu - Bengali Hallucination Detection | *Python, RAG, BanglaBERT, scikit-learn* GitHub
- Built a retrieval-augmented pipeline to verify Bengali answers against supplied context or a Bengali QA knowledge base.
- Created a hallucination dataset and used fine-tuned BanglaBERT with answer-comparison features to classify responses.

Redrick Routinson - Campus Automation | *Python, React, Flask, Firebase* GitHub Demo
- Built genetic-algorithm scheduling optimization, QR exam attendance, class swapping, events, forums, and lost-and-found.

Drone Vision - Aerial Tracking | *PyTorch, YOLOv11X, ByteTrack* GitHub
- Fine-tuned YOLOv11X on VisDrone and used ByteTrack for human counting and multi-object tracking.

Bengali Speaker Diarization - Speaker Turn Detection | *PyTorch, Silero VAD, ECAPA-TDNN* GitHub
- Built a speech pipeline using neural voice activity detection, speaker embeddings, and agglomerative clustering to label timestamped speaker turns.
- Estimated speaker counts with silhouette scores and exported diarization results to CSV.

Technical Skills

Programming Languages: Python, SQL, C++, Java
Machine Learning and Data: PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, feature engineering, model evaluation, forecasting, clustering and segmentation
Search, Ranking and Optimization: Hybrid BM25 + dense retrieval, reranking, information retrieval, genetic algorithms
Generative AI and NLP: RAG, LLMs, LangChain, LlamaIndex, Ollama, embeddings, BanglaBERT fine-tuning
Computer Vision and Audio: YOLOv11, ByteTrack, Silero VAD, ECAPA-TDNN, speaker diarization
Backend and APIs: REST APIs, Flask, Spring Boot, Node.js, RabbitMQ
Databases: PostgreSQL, MySQL, Firebase, ChromaDB (vector DB)
DevOps and Tools: Docker, Git, Linux
Web and Mobile: React, Next.js, Tailwind CSS, React Native, Flutter

Achievements and Awards

- **5th Place, VAND4 International Competition** — Zero-Shot Industrial Track **2026**
- **First Runner-up, IUT Coderush Hackathon** — Rapid team application development **2025**
- **7th Place, BAIC Datathon** — Data preprocessing, visualization, and modeling **2025**